BM20BC module

File: LeoPatch_MCU.kicad_sch

Power

File: LeoPatch_Power.kicad_sch

USB-C

File: LeoPatch_USB.kicad_sch

ECG front end

File: LeoPatch_ECG.kicad_sch

LEDs + button

File: LeoPatch_UI.kicad_sch

4xAA -> on/off -> reverse-polarity diode -> MCP1703A 3.3 V LDO -> BM20BC + AD8232 (USB-C 5 V also feeds the LDO)

ECG electrodes -> 3.5 mm jack -> AD8232 -> ECG_OUT -> nRF54LM20B AIN4 (P1.06)

Result LEDs, EXPORT button, SWD header (first flash), USB-C (updates + data). 4-layer, standard vias.

Rice ELEC 494 — LeoPatch

Sheet: /
File: LeoPatch_v2.kicad_sch

Title: LeoPatch v2 — newborn CCHD ECG screening (BM20BC, 4-layer, USB-

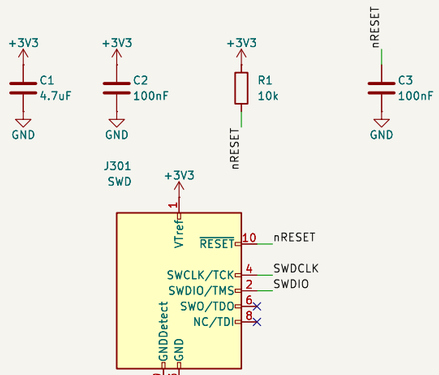
Size: A4	Date: 2026-11-03	**Rev: 2.0**
KiCad E.D.A. 9.0.7		Id: 1/6

MCU: Fanstel BM20BC module — nRF54LM20B (Arm M33 + Axon NPU), 32 MHz + 32 kHz crystals, DC/DC, chip antenna

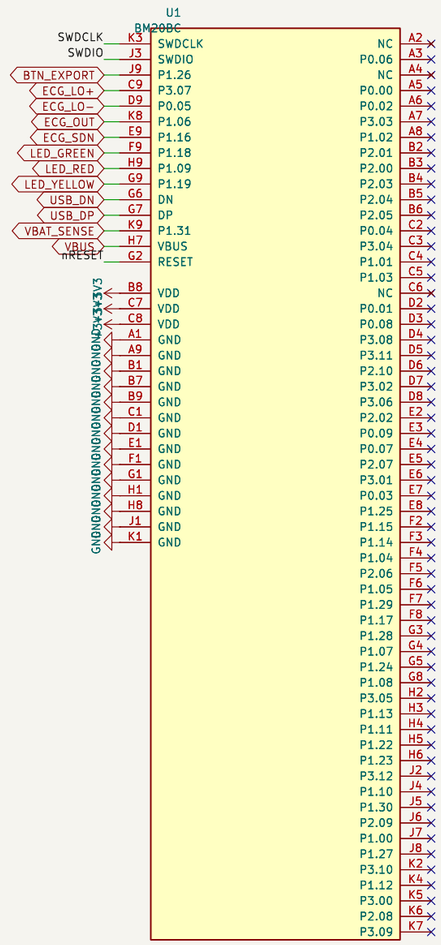
Antenna end of the module sits at the board edge over a copper keep-out (Fanstel mounting guideline).

USB: VBUS (H7), D− (G6), D+ (G7) go to the USB-C sheet. SWDIO, RESET and USB use vias between module pads.

Decoupling, reset, SWD (as Fanstel EV-BM20)



First flash only (MCUboot bootloader) with the nRF54LM20 DK "Debug out" 10-pin cable; after that, update over USB-C or BLE.



Rice ELEC 494 — LeoPatch

Sheet: /BM20BC module/

File: LeoPatch_MCU.kicad_sch

Title: LeoPatch v2 — Fanstel BM20BC (nRF54LM20B + NPU)

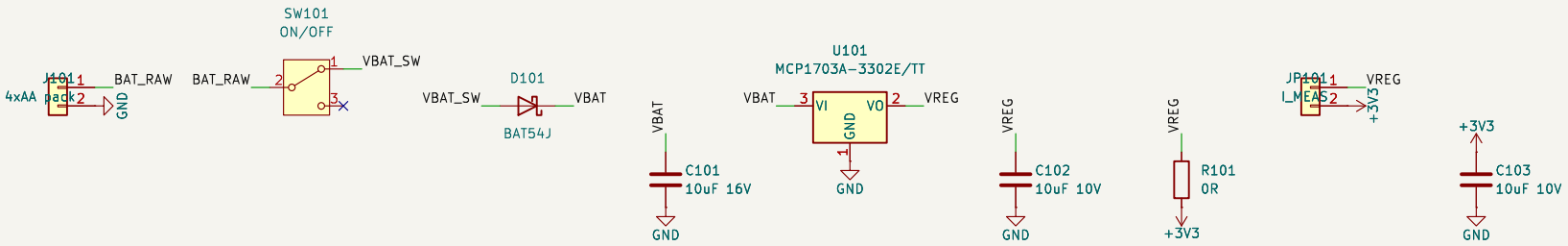
Size: A3	Date: 2026-10-25	Rev: 2.0
KiCad E.D.A. 9.0.7		Id: 2/6

POWER: 4xAA pack -> on/off switch -> reverse-polarity diode -> 3.3 V LDO

4xAA = 3.6 V (depleted) .. 6.4 V (fresh alkaline). MCP1703A: 16 V max in, 2 uA Iq, 250 mA.

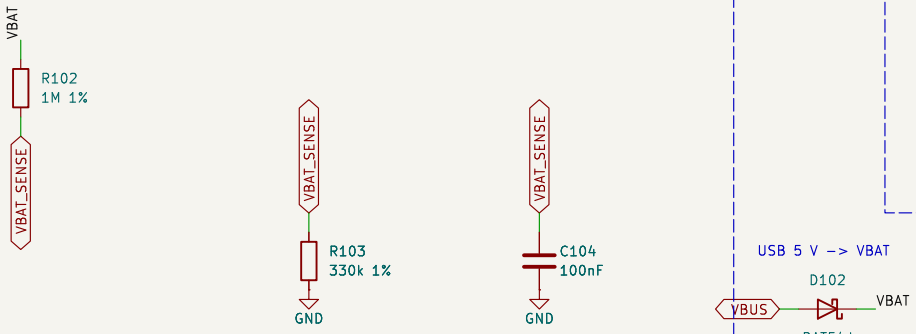
USB-C 5 V joins VBAT through D102 (diode-OR with the battery path); D101 blocks it from the cells.

Battery input + regulator



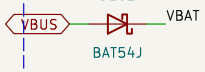
Current measurement: remove R101, connect PPK2 across JP101.

Battery voltage monitor -> BM20BC K9 P1.31 (AIN1)

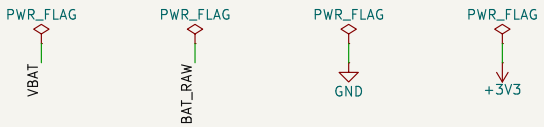


VBAT_SENSE = VBAT x 0.248 (6.4 V -> 1.59 V). ~4.8 uA divider load.

USB 5 V -> VBAT



ERC power flags



Rice ELEC 494 – LeoPatch

Sheet: /Power/

File: LeoPatch_Power.kicad_sch

Title: LeoPatch – Power (4xAA -> 3.3 V)

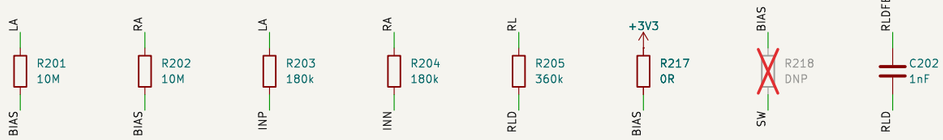
Size: A4	Date: 2026-10-25	**Rev: 2.0**
KiCad E.D.A. 9.0.7		Id: 3/6

ECG FRONT END: AD8232 — component values copied 1:1 from SparkFun AD8232 Heart Rate Monitor (CC BY-SA 4.0)

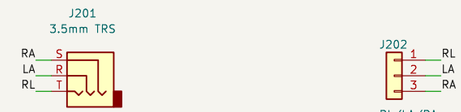
Same circuit as the red AD8232 breakout modules the team is testing; their on-board heartbeat LED (D1/R16/SJ4) is omitted.

Lead cable: 3.5 mm TRS plug, Tip = RL (yellow), Ring = LA (green), Sleeve = RA (red) — same as module.

Input bias / protection / right-leg drive



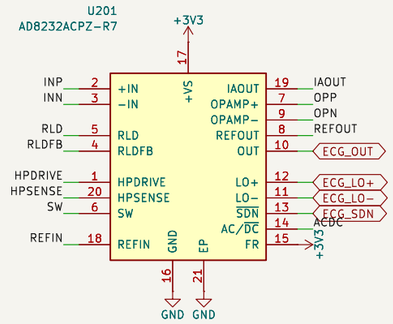
Electrode connectors



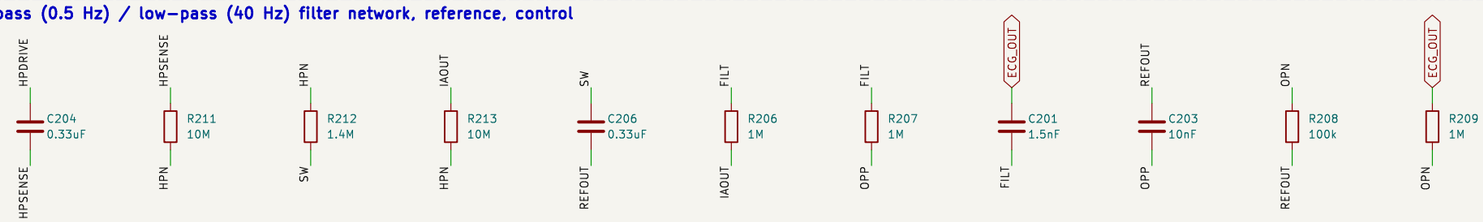


J202 = backup solder pads (same as module RA/LA/RL holes)

BM20BC pins: ECG_OUT -> K8 P1.06 (AIN4) LO+ -> C9 P3.07 LO- -> D9 P0.05 SDN -> E9 P1.16 (10k pull-up = always on)



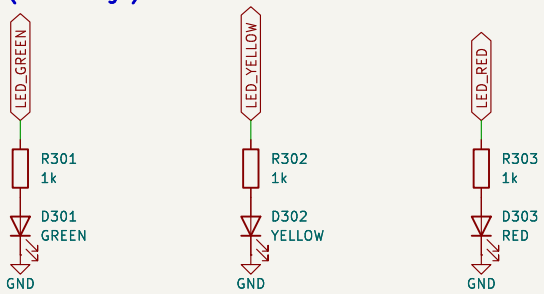
High-pass (0.5 Hz) / low-pass (40 Hz) filter network, reference, control



R219 fitted = AC lead-off mode (as SparkFun). R217 fitted / R218 DNP = inputs biased as SparkFun.

Rice ELEC 494 — LeoPatch		
Sheet: /ECG front end/		
File: LeoPatch_ECG.kicad_sch		
Title: LeoPatch — ECG front end (AD8232, SparkFun SEN-12650 circuit)		
Size: A3	Date: 2026-10-25	**Rev: 2.0**
KiCad E.D.A. 9.0.7		Id: 4/6

Result LEDs (active high)



Green = signal OK / normal, Yellow = needs review, Red = abnormal / bad signal

Export button



Export the recording over BLE (P1.26, internal pull-up)

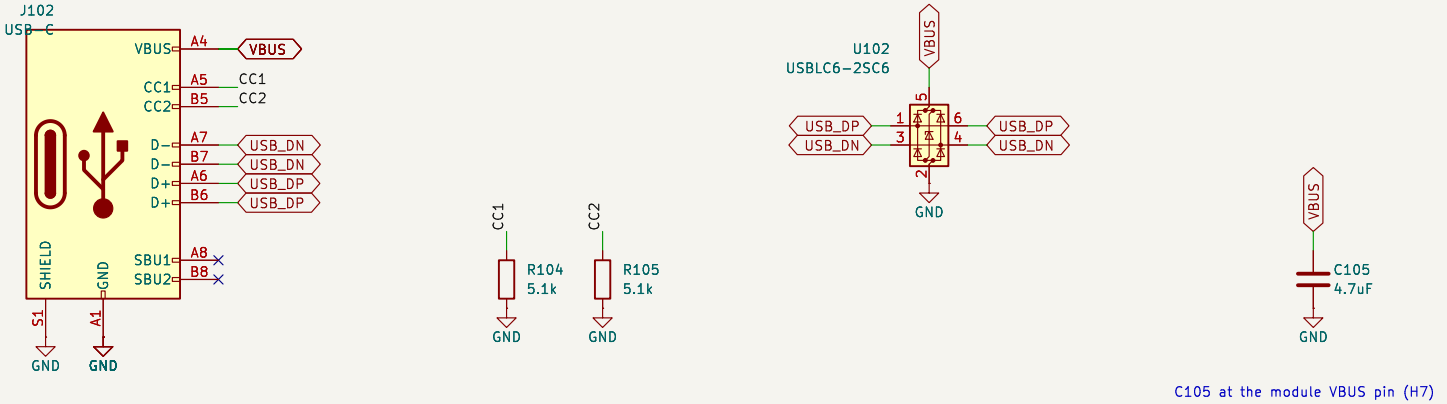
Rice ELEC 494 — LeoPatch		
Sheet: /LEDs + button/		
File: LeoPatch_UI.kicad_sch		
Title: LeoPatch v2 — Result LEDs and export button		
Size: A4	Date: 2026-10-25	**Rev: 2.0**
KiCad E.D.A. 9.0.7		Id: 5/6

USB-C: USB 2.0 device to the nRF54LM20B USB controller, plus 5 V into the regulator

5.1k on CC1/CC2 = sink (device). USBLC6-2SC6 ESD on D+/D-/VBUS, right at the connector.

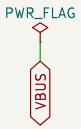
SAFETY: do not plug into a mains-powered computer while electrodes are on a patient (IEC 60601).

Connector, CC pull-downs, ESD, VBUS



C105 at the module VBUS pin (H7)

ERC power flag



Rice ELEC 494 – LeoPatch		
Sheet: /USB-C/		
File: LeoPatch_USB.kicad_sch		
Title: LeoPatch v2 – USB-C (update, data export, 5 V power)		
Size: A4	Date: 2026-10-25	**Rev: 2.0**
KiCad E.D.A. 9.0.7		Id: 6/6